

Design a Notification System - Step 1: Understand the Problem and Establish Design Scope

Introduction

Designing a notification system capable of sending **millions of notifications daily** is a complex system design problem.

The system must ensure:

- Scalability
- Reliability
- Low latency
- Multi-platform support

Requirement Clarification (Candidate ↔ Interviewer Discussion)

Question 1: What notification types are supported?

Interviewer:

- Push notifications
- SMS messages
- Emails

Question 2: Is this a real-time system?

Interviewer:

- It is a **soft real-time system**
- Notifications should be delivered:
 - As quickly as possible
- Small delays are acceptable during:
 - High traffic
 - Heavy workload situations

Question 3: What devices are supported?

Interviewer:

- iOS devices
- Android devices
- Laptop/Desktop systems

Question 4: What triggers notifications?

Interviewer:

Notifications can be triggered by:

1. Client Applications

Examples:

- User messages
- Likes/comments
- App events

2. Server-side Scheduled Jobs

Examples:

- Daily reports
- Promotional campaigns
- Reminder notifications

Question 5: Can users opt out?

Interviewer:

- Yes
- Users can disable notifications
- Opted-out users must not receive messages

Question 6: What is the system scale?

Interviewer:

Daily notification volume:

- **10 million** mobile push notifications
- **1 million** SMS messages
- **5 million** emails

Key Requirements Identified

Functional Requirements

1. Support Multiple Notification Types

- Push notifications
- SMS
- Emails

2. Multi-device Support

- Mobile devices
- Web/Desktop systems

3. Trigger-based Notifications

- User-generated events
- Scheduled system events

4. User Preference Management

- Opt-in / Opt-out support

Non-Functional Requirements

1. Scalability

- Handle millions of notifications daily

2. Reliability

- Notifications should not be lost

3. Low Latency

- Notifications delivered quickly

4. Fault Tolerance

- System should continue working during failures

5. Flexibility

- Easy to add new notification channels in future

Notification Volume Estimation

Notification Type	Daily Volume
Push Notifications	10 Million
SMS Messages	1 Million
Emails	5 Million

Key Takeaway

The notification system must be:

- Highly scalable
- Reliable
- Multi-channel
- Multi-device compatible

Core supported channels:

- Push Notifications
- SMS
- Emails

System should support:

- Real-time delivery
- User preferences
- Massive scale